

CentOS DHCP and DNS Server Administration Lab

Project Overview

Project Overview

Project Title	CentOS DHCP and DNS Server Administration Lab
Project Category	Linux Server Administration / Network Services
Platform	VMware Workstation
Operating System	CentOS Linux
Network	VMware VMnet1 Host-Only Network
Network Address	192.168.10.0/24
Server IP Address	192.168.10.2
DNS Domain	lab.local

1. Introduction

This project demonstrates the deployment and administration of a CentOS-based network services server in a virtualized VMware environment.

The main purpose of the project is to configure a CentOS server as a centralized DHCP and DNS server and connect multiple CentOS client machines to the server.

The DHCP service automatically provides IP addresses and network configuration to client systems. The DNS service provides hostname resolution within the lab.local domain.

The project also demonstrates the integration of DHCP and DNS services, client configuration, network testing, troubleshooting, and technical documentation.

2. Project Environment

The project was developed using VMware Workstation and CentOS Linux virtual machines.

The environment consists of:

- One CentOS Server
- Two CentOS Client Machines
- VMware VMnet1 Host-Only Network

The CentOS server provides:

- DHCP service
- DNS service
- Forward DNS resolution
- Reverse DNS resolution

The clients obtain their network configuration automatically using DHCP.

3. Network Architecture

The network uses the 192.168.10.0/24 private network.

The server is configured with a static IP address: 192.168.10.2

The server provides DHCP and DNS services to the client systems.

The clients receive their IP addresses dynamically from the DHCP server and use the server at 192.168.10.2 for DNS resolution.

4. Main Services

DHCP Server

The DHCP service automatically assigns IP addresses and network configuration to client machines.

The DHCP configuration provides:

- IP address
- Subnet mask
- Default gateway
- DNS server address
- DNS domain

DNS Server

The DNS service is configured using BIND.

The DNS server provides hostname resolution for the lab.local domain.

The project includes:

- Forward DNS zone
- Reverse DNS zone
- A records
- PTR records

5. DHCP and DNS Integration

The DHCP server is configured to provide the DNS server address 192.168.10.2 to clients.

When a client receives an IP address through DHCP, it also receives the DNS server information.

This allows clients to automatically use the internal DNS server for hostname resolution.

6. Client Systems

Two CentOS client systems are used in the project.

Client 1

MAC Address: 00:0c:29:97:b6:5b

Client 2

MAC Address: 00:0c:29:1e:f9:43

Both clients are configured to obtain their IP addresses automatically through DHCP.

7. Testing

The following tests were performed:

- Server-to-client ping
- Client-to-server ping
- Client-to-client connectivity
- DHCP IP assignment
- DHCP lease verification
- DNS service verification
- Forward DNS lookup
- Reverse DNS lookup
- Hostname resolution
- DHCP-DNS integration testing

8. Troubleshooting

During the project, several network and service-related issues were investigated.

Troubleshooting activities included:

- Checking network interfaces
- Checking VMware network configuration
- Checking IP configuration
- Checking routing tables
- Checking DHCP service status
- Checking DHCP logs
- Checking DHCP leases
- Checking DNS service status
- Validating DNS configuration
- Testing DNS port 53
- Testing DHCP port 67
- Checking firewall configuration
- Troubleshooting the default virbr0 interface

9. Project Outcome

The project successfully demonstrates the deployment and administration of centralized DHCP and DNS services on CentOS Linux.

The DHCP server successfully assigns network configuration to client systems.

The DNS server successfully resolves hostnames and IP addresses using forward and reverse DNS.

The project provides practical experience in Linux server administration, network services, virtualization, troubleshooting, and technical documentation.